

Beveridge Curve in the Dynamic Model

Pascal Michailat
<https://pascalmichailat.org/w/>

$$\begin{cases} \theta^* = 1 \\ u^* = \sqrt{uv} \end{cases} \text{ requires } \begin{cases} \rho = 1 \text{ (recruiting cost)} \\ \text{Beveridge curve is hyperbola.} \end{cases}$$

Beveridge curve in dynamic model:

$$u(\theta) = \frac{\lambda}{\lambda + f(\theta)}$$

$$\Rightarrow u = \frac{\lambda}{\lambda + \mu [v/u]^{1-\eta}}$$

job-separation rate
matching elasticity

$$\Rightarrow \lambda u + \mu v^{1-\eta} u^\eta = \lambda$$

$$\Rightarrow \mu v^{1-\eta} u^\eta = \lambda(1-u)$$

$$\Rightarrow v^{1-\eta} = \frac{\lambda(1-u)}{\mu u^\eta}$$

$$\Rightarrow v(u) = \left[\frac{\lambda(1-u)}{\mu u^\eta} \right]^{1/(1-\eta)}$$

$$\frac{dv}{du} = \frac{1}{1-\eta} \left[-\frac{u}{1-u} - \eta \right] = -\frac{1}{1-\eta} \left[\eta + \frac{u}{1-u} \right]$$

$$\eta \gg u/(1-u)$$

so Beveridge curve is
almost isoelastic

$$\begin{cases} u \sim 5\% \\ 1-u \sim 0.95\% \\ u/(1-u) \sim 5\% \end{cases}$$

$\eta \sim 0.5$

If $\eta = 0.5$ (Petrangolo & Pissavide 2001)
then $\frac{d\ln \sigma}{d\ln u} = -2 \left(0.5 + u/(1-u) \right) = -\left(1 + u/(1-u) \right)$

$$\frac{d\ln \sigma}{d\ln u} \approx -1$$

To have hyperbola at efficiency point.

$$\text{set } \eta \text{ such that } \frac{\eta}{1-\eta} + \frac{u^*}{(1-\eta)(1-u^*)} = 1$$

$$u^* = u^0 \rightarrow \frac{\eta}{1-\eta} + \frac{0.04}{(1-\eta) \times 0.96} = 1$$

$$\rightarrow \eta \approx 0.44$$